How-To Library

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Prerequisites to configure cross-account access to Amazon S3 buckets using IAM AssumeRole in Databricks

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Abstract

This article explains the prerequisite tasks that you must complete before you configure cross-account access to Amazon S3 buckets using IAM AssumeRole in Databricks.

Supported Versions

- Informatica Cloud® Data Integration Databricks Connector

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Overview

You can configure cross-account access to Amazon S3 buckets using IAM AssumeRole in Databricks.

Let's understand how to enable cross-account access with the help of an example.

Consider an AWS Account A with account ID <databricks-ws-acct-id> and an AWS Account B with account ID <bucket-ec2-acct-id>.

- Account A is the one you used to sign up for Databricks and hosts the Databricks workspace. The EC2 service and DBFS root bucket are managed by this account.
- Account B owns the S3 bucket <s3-bucket-name-in-AccB> and hosts the Secure Agent.

To enable cross-account access, configure Account A to use the AWS AssumeRole to access files on Account B's role. Enabling this access requires you to configure in Account A, Account B, and the Databricks Admin Console.

This article describes the steps to configure account A to use the AWS AssumeRole action to access files on <s3-bucket-name-in-AccB> as Account B's role.

You also need to configure the Databricks cluster and SQL warehouse to access the Amazon S3 bucket.

Account configurations

Configure the roles and policies for Account A and Account B.

Create IAM role in Account A and attach the policy

Perform the following steps to configure the settings for Account A:

1. Create a role called RoleA in Account A.
2. Enter the ARN of the instance profile as `arn:aws:iam::<databricks-ws-acct-id>:instance-profile/RoleA` to assume a role to RoleA.
3. Create a policy that allows a role in Account A to request temporary credentials for a role in Account B and attach that policy to RoleA.
   Sample policy to show how IAM role in Account A fetches temporary credentials from RoleB in Account B:

```
{
  "Version": "2023-09-20",
  "Statement": [
    {
      "Sid": "Stmt14878840014322",
      "Effect": "Allow",
      "Action": [
        "sts:AssumeRole"
      ],
      "Resource": [
        "arn:aws:iam::<databricks-ws-acct-id>:bucket-ec2-acct-id>:role/RoleB"
      ]
    }
  ]
}
```

Allow PassRole permission to RoleA on an IAM role

You can enable AssumeRole permission to RoleA by providing the policy for PassRole in the Databricks console.

Perform the following steps to enable PassRole permissions to RoleA of Account A:

1. Go to the **Databricks console > Workspaces > Credentials**
2. Append the policy `iam:PassRole` to the end of the Statement array on the policy attached to the IAM role.

Sample policy to show PassRole permissions of an IAM role assigned to RoleA:

```
{
  "Effect": "Allow",
  "Action": "iam:PassRole",
  "Resource": "arn:aws:iam::<databricks-ws-acct-id>:role/RoleA"
}
```

Note: Don't overwrite any existing policy that was created in the Databricks deployment.

Create IAM role in Account B and attach the policy

Perform the following steps to configure the settings for Account B:

1. Create a role called RoleB in Account B where the ARN of RoleB is `arn:aws:iam::<bucket-ec2-acct-id>:role/RoleB`.
2. Create a policy that allows a role in Account B to access the required S3 bucket.
3. Attach the policy to RoleB.

```
{
    "Version": "2023-09-20",
    "Statement": [
        {
        "Sid": "VisualEditor1",
        "Effect": "Allow",
        "Action": "s3:*",
```

```
            "Resource": [
                "arn:aws:s3:::<s3-bucket-name-in-AccB>/*",
                "arn:aws:s3:::<s3-bucket-name-in-AccB>"
            ]
            }
        ]
        }
```

4. Once you edit the trust relationship, Account A's role RoleA can assume Bccount B's role. To edit the trust relationship, go to **Select IAM > Roles > RoleB > Trust relationships > Edit trust relationship**.

Sample policy for Account A's role to assume Account B's role using `sts:AssumeRole`:

```
{
    "Version": "2023-09-20",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": {
                "Service": "ec2.amazonaws.com"
            },
            "Action": "sts:AssumeRole"
        },
        {
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::<databricks-ws-acct-id>:role/RoleA"
            },
            "Action": "sts:AssumeRole"
        }
    ]
}
```

Create a bucket policy for the S3 bucket in Account B

Perform the following steps to configure the S3 bucket policy in Account B:

1. To configure the S3 bucket policy, go to **S3 > <s3-bucket-name> > Permissions > Bucket Policy** in the AWS console.
2. Include RoleB in the bucket policy.
   The following snippet shows the list of actions that are performed on the S3 bucket:

```
    {
        "Version": "2012-10-17",
        "Statement": [
            {
                "Sid": "Example permissions",
                "Effect": "Allow",
                "Principal": {
                    "AWS": " arn:aws:iam::<bucket-ec2-acct-id>:role/RoleB"
                },
                "Action": [
                    "s3:GetBucketLocation",
                    "s3:ListBucket"
                ],
                "Resource": "arn:aws:s3:::<s3-bucket-name-in-AccB>"
            },
            {
                "Effect": "Allow",
                "Principal": {
                    "AWS": " arn:aws:iam::<bucket-ec2-acct-id>:role/RoleB"
                },
                "Action": [
                    "s3:PutObject",
                    "s3:GetObject",
                    "s3:DeleteObject",
                    "s3:PutObjectAcl"
                ],
```

```
                "Resource": "arn:aws:s3:::<s3-bucket-name-in-AccB>/*"
            }
        ]
    }
```

Cluster configurations

Create the instance profile and configure the Databricks cluster and SQL warehouse to use the instance profile.

Create the instance profile in Databricks

To assume a Role in AWS console across different accounts, configure the account settings for the IAM role in the Databricks instance profile.

Perform the following steps to configure a Databricks instance profile:

1. Go to the **Admin Settings> Instance Profile> Add Instance profile**.
2. Copy and paste the instance profile ARN to the **Instance profile ARN field**: `arn:aws:iam::<databricks-ws-acct-id>:instance-profile/RoleA`.
3. Copy and paste the role ARN into the **IAM Role ARN field**: `arn:aws:iam::<databricks-ws-acct-id>:role/RoleA`.
4. Name the instance profile and select the **Type** as `Regular instance profile`.
5. Configure the access controls for the required users and groups for the instance profile.
6. Save the instance profile.

Configure Databricks cluster to use instance profile

After you create an instance profile, configure the Databricks cluster to use the new instance profile.

Perform the following steps to configure the Databricks cluster:

1. Create a new Databricks cluster or select an existing cluster from the `Cluster settings` page.
2. From the advanced options section, select the new instance profile you had created from the `Instance profile` tab.
3. In the Spark configurations tab, specify the following AssumeRole authentication type properties and the ARN to delegate the role to RoleB.
   a. `fs.s3a.bucket.<bucket-ec2-acct-id>.aws.credentials.provider` *`org.apache.hadoop.fs.s3a.auth.AssumedRoleCredentialProvider`*
   b. `fs.s3a.bucket.<bucket-ec2-acct-id>.assumed.role.arn` *`arn:aws:iam::<bucket-ec2-acct-id>:role/RoleB`*

Configure SQL warehouse to use instance profile

After creating an instance profile you must also configure the SQL warehouse to use the new instance profile.

Perform the following steps to configure the SQL warehouse to the new instance profile:

1. Go to **Admin settings > SQL warehouse settings > Instance profile** in the Admin panel.
2. Select the new instance profile you had created from the `Instance profile` tab.

3. Under SQL warehouse settings, go to `Data Access configuration` page and add the following properties:
   a. `spark.hadoop.fs.s3a.bucket.<bucket-ec2-acct-id>.aws.credentials.provider` *`org.apache.hadoop.fs.s3a.auth.AssumedRoleCredentialProvider`*.
   b. `spark.hadoop.fs.s3a.bucket.<bucket-ec2-acct-id>.assumed.role.arn` *`arn:aws:iam::<bucket-ec2-acct-id>:role/RoleB`*.
4. Save the cluster settings.

Configuring Databricks accounts with EC2 role using EC2 instance identifier

Attach RoleB to EC2 security and Add sts:assumerole with EC2 instance identifier to request credentials.

Attach RoleB to EC2 security

After the role and cluster configurations, configure the EC2 role for the AssumeRole.

Perform the following steps to configure the EC2 role to the AssumeRole for RoleB:

1. Go to **Actions > Security > Modify IAM role** on the EC2 node where the Secure Agent is present.
2. Select the `arn:aws:iam::<bucket-ec2-acct-id>:role/RoleB`.

Add sts:assumerole with EC2 instance identifier to request credentials

After you configure the EC2 role, add the sts:assumerole with EC2 instance identifier to request the temporary credentials to stage data in the S3 bucket.

In order to do so, modify the AssumeRole policy attached to RoleA in Account A and add the ARN `arn:aws:sts::databricks-ws-acct-id:assumed-role/RoleB/<ec2InstanceID>` in the resource field.

The following sample policy shows the sts:assumerole with EC2 instance identifier with the role ARN:

```
{
  "Version": "2023-09-20",
  "Statement": [
    {
      "Sid": "Stmt14878840014322",
      "Effect": "Allow",
      "Action": [
        "sts:AssumeRole"
      ],
      "Resource": [
        "arn:aws:iam::<bucket-ec2-acct-id>:role/RoleB",
        "arn:aws:sts::databricks-ws-acct-id:assumed-role/RoleB/<ec2InstanceID>"
      ]
    }
  ]
}
```

Author

Adrija Pandya

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